Phase 1 Project Proposal (Updated Implementation Version)

Pakistan Weather Analytics Lakehouse Pipeline

Topic: Pakistan Weather Analytics Lakehouse Pipeline (Lahore, Karachi, Islamabad, Peshawar, Quetta)
Course: Data Analysis and Visualization (BDS - 5B)
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Domain: Climate & Meteorological Analytics
Data Architecture: Medallion Architecture (Bronze → Silver → Gold) via Apache Spark
GitHub Repository:
https://github.com/Syedafaryalzehra/Lahore-Weather-Analytics-Lakehouse-Pipeline/tree/main

1. Domain & Source Identification

Project Concept

This project builds an automated, distributed data pipeline for weather and climate analytics across five major cities of Pakistan: Lahore, Karachi, Islamabad, Peshawar, and Quetta. Weather data is one of the most consistently generated, publicly available, and analytically rich datasets in the world. The five cities cover distinct climate zones: the coastal port of Karachi, the plains cities of Lahore and Peshawar, Islamabad at the foot of the Margalla Hills, and high-altitude Quetta. By ingesting over 26 years of meteorological readings for each city, the pipeline enables analysts, urban planners, and researchers to identify long-term climate trends, seasonal precipitation patterns, and extreme weather variations across regions.

Data Source

We utilize the Open-Meteo Historical Weather API, an open-source meteorological API requiring no API key, registration, or authentication.

- **API Name:** Open-Meteo Archive API
- **Base URL:** https://archive-api.open-meteo.com/v1/archive
- **Data Format:** JSON responses containing weather measurement arrays.
- **Source Access & Cost:** Fully public, $0.00 cost. Fair-use rate limits are managed via retries and exponential back-off in extraction notebooks.
- **Variables:** 42 variables per city covering temperature, humidity, pressure, precipitation, wind speed, cloud cover, and radiation metrics.

Target City Locations

City	Requested Latitude	Requested Longitude
Lahore	31.5204	74.3587
Karachi	24.8607	67.0011
Islamabad	33.6844	73.0479
Peshawar	34.0151	71.5249
Quetta	30.1798	66.9750

Ingestion Pattern & Dual-Load Capability

- **Full Load Baseline:** Extracted using start_date=2000-01-01 and end_date=2026-09-30 across 6 historical date chunks per city (30 API payloads total) merged into pakistan_weather_fullload.json. This establishes a baseline of 1,172,400 hourly records (234,480 per city).
- **Incremental Load:** Scheduled incremental runs fetch only uncollected days up to the most recent finalized archive date, appending raw JSON files into Bronze and upserting cleaned hourly rows into Silver. A daily run appends 120 hourly records (5 cities × 24 hours).
- **Backfill Capability:** Parameterized notebooks accept date range, batch ID, and load type parameters to reprocess any target period without code modifications.

2. Data Samples & Volume

Sample Files on GitHub

- data/full_load/full_load_sample.json: 7 days of sample hourly weather data (Jan 1–7, 2024) across all 5 cities.
- data/incremental_load/incremental_load_sample.json: 1 day of incremental hourly readings (Jan 8, 2024) across all 5 cities.

Volume & Row Count Summary

Pipeline Layer / Table	Granularity	Expected Row Count	Data Footprint
Raw Ingestion Payload	Merged API JSON responses	30 payloads (Full) + 1 / daily run	~240 MB JSON raw baseline
Bronze Layer (bronze_weather_raw)	1 row per city per chunk response	30 rows (Full baseline)	Delta format with raw arrays intact
Silver Layer (silver_weather_hourly)	1 row per city per hour	1,172,400 rows (Baseline)	~25–30 MB compressed Delta
Gold Fact (fact_weather_daily)	1 row per city per calendar day	48,850 rows (Jan 2000 – Sept 2026)	< 2 MB compressed Delta

3. Security & Governance

PII Identification & Governance

- **PII Status:** Non-PII dataset. Contains solely environmental sensor measurements, standardized timestamps, and fixed city coordinates.
- **Timestamp Standardization:** Stored and processed in GMT/UTC to align all 5 cities on a single temporal basis. Local Pakistan Standard Time (UTC+5) is derived in reporting.
- **Schema Enforcement:** Explicit StructType reading in Bronze with a _rescued_data column to prevent missing columns or silent data loss.
- **Data Quality Quarantine:** Out-of-bounds readings (temperatures outside -30 °C to 55 °C, negative precipitation, or invalid timestamps) are routed to silver_quarantine with rejection audit reasons rather than silently dropped.

4. Medallion Data Modeling & Architecture

`[ Open-Meteo REST API ] → [ Bronze Delta ] → [ Silver Delta ] → [ Gold Star Schema ] → [ Power BI ]`

Layer Specifications

- **Bronze Layer (bronze_weather_raw):** Stores raw JSON API responses as Delta rows with nested measurement arrays. Appends lineage metadata (load_timestamp, load_type, batch_id, source_file). Every run writes audit records to pipeline_execution_logs.
- **Silver Layer (silver_weather_hourly):** Explodes hourly arrays into 1 row per city per hour (1,172,400 records). Applies explicit data casting, handles isolated nulls, extracts calendar attributes, and performs idempotent upserts (MERGE INTO) on (observation_timestamp, latitude, longitude).
- **Gold Layer (Star Schema Architecture):**
    - **dim_location:** Dimension table holding city attributes (location_key, city, your requested/grid coordinates, and province).
    - **dim_date:** Dimension table holding calendar attributes across 2000–2030 (11,323 rows; date_key, full_date, year, month, month_name, day, day_name, season, is_monsoon, is_weekend).
    - **fact_weather_daily:** Central fact table aggregated to 1 row per city per day (48,850 rows). Holds daily temperature averages/extremes, total precipitation, and max wind speed linked via location_key and date_key.

5. Business Intelligence & Dashboard Implementation

The Power BI dashboard operates on a Star Schema model connected to Databricks Gold tables via Import Mode. It comprises interactive global slicers (city and year) and the following visuals:

Dashboard Page 1: Weather Analytics & Trends

- **Geographic Map Visual:** Plotted using city_latitude and city_longitude with bubble size representing Total Precipitation across target cities.
- **Average Monthly Rain Bar Chart:** Clustered column visual displaying average precipitation (mm) grouped by month_name and color-coded by season (Autumn, Spring,

Summer, Winter) to highlight monsoon rainfall spikes (July–August).
- **Monthly Temperature Matrix Heatmap:** Matrix visual displaying monthly average temperatures across years (2000–2026), applying conditional color formatting (blue for cold to deep red for extreme heat) to spot multi-decade warming trends and monthly anomalies.
- **Annual Average Temperature Line Chart:** Multi-line chart tracking Avg Temperature C by year and city across all 5 Pakistani cities over the 26-year historical timeline.

Dashboard Page 2: Pipeline Operations & FinOps Audit

- **Pipeline Performance Bar Chart:** Visualizing row throughput across pipeline layers (Bronze, Silver, Gold) extracted from pipeline_execution_logs.
- **Execution Audit Table:** Tabular display showing run statuses, duration metrics, and batch identifiers for operational monitoring.

6. Engineering Setup & FinOps

- **Infrastructure:** Databricks Community Edition (Apache Spark engine).
- **Storage & Compression:** Delta Lake format with Snappy compression and Liquid Clustering on Gold tables.
- **Total Project Infrastructure Cost:** $0.00 (Fully realized on free tier compute, storage, and visualization platforms).